SIMON HALVDANSSON

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EDUCATION

NTNU Norwegian University of Science and Technology Aug 2021 - Jun 2025
Ph.D. in Mathematics
Thesis: Extensions of Quantum Harmonic Analysis and Applications to Time-Frequency Analysis

Lund University Aug 2016 - Jun 2021
M.Sc. and B.Sc. in Mathematics GPA: 4.0/4.0
Master thesis: Existence and Approximations of Optimal Wavelets
Bachelor thesis: Computations with the 2D Coulomb Gas

EXPERIENCE

SINTEF Energy Research May 2025 - Present
Research Scientist *Trondheim, Norway*

- Worked on forecasting of electricity reserve markets
- Developed a system for LLM-grading of database events with structured outputs, ollama, and the Gemini API
- Applied model predictive control and machine learning for efficient control of integrated energy systems
- Project development, writing grant applications

NTNU Norwegian University of Science and Technology Aug 2021 - May 2025
PhD Candidate *Trondheim, Norway*

- Authored 8 research articles in the fields of time-frequency analysis, functional analysis and operator theory
- Presented my work to international audiences at 10+ conferences, workshops and seminars
- Held exercise sessions, wrote problems and solutions, mentored students and graded exams (2 courses per year)
- Refereed papers for JFAA, JMP, SampTA, JFA, JGA among others

RSA Ductor Summer 2017
Web Developer Consultant *Stockholm, Sweden*

- Developed a web application to support internal business processes using AngularJS

DeLaval Summer 2015 and 2016
Web Developer *Stockholm, Sweden*

- Contributed to design and development of mobile web application and development of Java backend

PROJECTS

Android apps

- Developed several Android applications with a combined total of 200 000+ downloads and €2500+ revenue
- Maintaining the most popular Hacker News client for Android, Harmonic, 7 000+ DAU, 900+ GitHub stars

Technical blog

- Writing a blog about technical problems in machine learning and time-frequency analysis

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, C++, C#, MATLAB
Libraries	PyTorch, NumPy, SciPy, Android, LTFAT, Matplotlib, Pyomo, SCIP, ollama